



Sources

do you see any geographic and/or strategic patterns of Ukrainian attacks against Russia's or occupied territories energy and/or military infrastructures in the last 4 months?

Here's a summary of observed geographic and strategic patterns in Ukrainian strikes on Russian and occupied-territory energy and military infrastructure over the past ~4 months (≈ Sept–Dec 2025) based on multiple verified reporting and analysis:

Recent Ukraine–Russia War Developments (energy/military strikes)



BI Business Insider

Ukrainian drones struck a Russian oil platform in the Caspian Sea for the third time in a matter of days, security source says

Yesterday



R Reuters

Ukraine says underwater drones hit submarine, but Moscow denies damage

Today



FT Financial Times

Ukraine claims strike on Russian submarine with underwater drones

Yesterday



📍 1) Deep-strike Campaign on Russian Energy Infrastructure

Caspian Sea oil platforms (Dec 2025)

- Ukrainian drones have repeatedly *struck Russian offshore oil platforms* in the Caspian Sea (e.g., Filanovsky, Korchagin) multiple times within days, **damaging production and forcing shutdowns**.
- These hits are part of a growing *long-range campaign aimed at undermining Russian oil revenue and logistics* rather than purely tactical battlefield goals. Business Insider

➡ **Pattern:** moving beyond border regions deeper into Russia's industrial infrastructure, using drones to target **oil extraction and export nodes**.

2) Novel Naval and Strategic Military Targets

Underwater drone strike on Novorossiysk submarine (Dec 15, 2025)

- Ukraine reportedly used domestically developed *underwater drones* to hit a Russian **Kilo-class submarine** at the Novorossiysk base — a *new form of asymmetric naval warfare*.
- If verified, this indicates *strategic targeting of high-value military assets*, not just energy infrastructure. Reuters +1

➡ **Pattern:** expanding targeting methods (underwater drones) and *high-value military hardware* beyond land and oil facilities.

3) Persistent Strikes on Occupied and Border Energy/Military Sites

Drone and missile strikes in occupied Donetsk & Novgorod/Smolensk regions

- Reports mention Ukrainian *drone strikes on chemical plants and infrastructure in occupied territories* (e.g., near Novgorod, Smolensk), which are tied to Russia's war production.
- Frontline drone strikes in Donetsk aim at *thermal power plants and substations*, sometimes causing localized blackouts. The Guardian

➡ **Pattern:** continued targeting of **occupied eastern infrastructure**, both *energy and war-production linked*, to degrade enemy sustainment.

4) Strategic Geographic Spread

From Western Russia's industrial heartlands to frontline areas

Analysis from think tanks and conflict monitors indicates:

- Ukraine has struck power substations and energy nodes deep in **western and central Russia** — e.g., near Volga River regions and industrial hubs (though precise reporting varies).
- Strikes also hit **fuel depots, refineries, and logistical hubs** beyond the immediate

frontlines (e.g., pipeline sections in occupied Crimea, refinery infrastructure in Siberia).

Chatham House +1

➡ **Pattern:** Combining **deep strategic impact strikes** (economically significant targets) with **theater-level operational hits** near occupied zones.

 5) Strategic Objectives Behind These Patterns

 Economic pressure

- Strikes on oil platforms, refineries, pipeline infrastructure and export shipping aim to **degrade Russia’s ability to fund and sustain the war** by targeting *fuel revenue flows*.

Chatham House

 Operational disruption

- Attacks on power substations, thermal plants and logistics infrastructure in occupied areas target **Russian frontline sustainment** and reduce war-production capacity.

The Guardian

 Psychological and deterrence impact

- Repeated long-range deep strikes (e.g., Caspian and Black Sea targets) signal Ukrainian reach and **challenge Russian assumptions of sanctuary for energy assets**.

 Overall Geographic/Strategic Pattern (last ~4 months)

Target Zone	Type of Target	Primary Strategic Goal
Deep inside Russia (Caspian, Volga, Siberian oil facilities)	Oil platforms, refineries, pipelines	Economic attrition & revenue disruption
Occupied East (Donetsk, Zaporizhzhia)	Energy grids, power plants, substations	Frontline sustainment disruption
Russian military naval bases (Novorossiysk)	Submarine and naval assets	“High-value military” degradation
Border regions & logistics nodes	Chemical plants, depots	Operational war-production impact

Emerging Strategic Shift

In 2025, especially in the last few months, Ukrainian strikes have **expanded beyond tactical battlefield support** to:

1. Strategic economic infrastructure deep within Russia
2. New domains of warfare (naval/underwater drones)
3. Target sets linked to both war financing and war production
4. Sustained pressure on occupied territory infrastructure

This contrasts with earlier phases where Ukrainian strikes focused more narrowly on **frontline support and tactical counter-battery operations**.

Contextual Note

Reliable independent verification of some deep-strike claims varies — especially Russian energy infrastructure impacts deep within Russia — due to reporting asymmetries and information security concerns. Analysts generally view the pattern as **escalatory but focused on strategic economic and sustainment targets**, not indiscriminate attacks on civilian infrastructure.



Sources

any patterns what explicitly refers to a preparation of some counterattack in some region?

Here's a **current, evidence-based look** at whether there are any **patterns in recent months that could indicate Ukrainian preparations for a counteroffensive** — rather than just reactive strikes — based on publicly available reporting and expert analysis.

Recent Ukraine-Russia war developments relating to military and strategic trends



 Reuters

European leaders agree
Ukraine security
guarantees should include
European-led
peacekeeping force

Today



 The Guardian

Europe ready to lead
'multinational force' in
Ukraine as part of US
peace plan

Today



 The Guardian

European leaders propose
'multinational force' and
hail 'significant progress'
on Ukraine talks - as it
happened

Today



1) Frontline dynamics — tactical pushes rather than a large counteroffensive... (Dec 2025)

✓ Reclaimed territory around Kupiansk

- Ukrainian forces recently recaptured much of Kupiansk in Kharkiv region after intense fighting, marking a notable local tactical success. Le Monde.fr

Interpretation

- This reflects **localized offensive action** aimed at improving tactical positions.
- It is *not necessarily a large strategic counteroffensive* — it's more of a focused push within a specific sector.

2) No clear signs of broad counteroffensive buildup in public sources

In recent Western reporting and expert assessments from groups like the Institute for the Study of War (ISW), the focus has been more on **Russian operations or negotiations**, not on a massive Ukrainian buildup:

- Analysts have warned that Russia might try to exploit ceasefires to regroup for its own offensives rather than Ukraine preparing a major counterattack. Pravda

Pattern takeaway

- No widely reported **large Ukrainian force build-up or redeployment** that would

clearly signal preparation for a major multi-front counteroffensive has been visible in the last ~4 months in open reporting.

3) Continued long-range strikes don't necessarily equal counteroffensive prep

Ukraine's ongoing use of drones and long-range weapons (e.g., strikes on infrastructure or strategic targets) has been widely reported.

While these indicate **pace and capability**, they don't necessarily become battlefield preparations for a ground offensive without:

- ground force concentration
- logistics build-up
- air support readiness
- sustained supply lines

Public sources **don't show clear widescale evidence** of all of these converging into open counteroffensive preparations — at least not in the past ~4 months.

4) Ongoing diplomacy and security guarantees shaping planning

European and U.S. diplomatic efforts are focusing on **security guarantees and peace plans** rather than an imminent offensive:

- A **European-backed multinational force** proposal is being explored as part of a peace process framework. Reuters +1
- Discussions over peace negotiations and territorial arrangements continue this month.

The Guardian

Strategic implication

- This may mean Kyiv is pursuing a **combined strategy** of defensive operations and international political leverage, rather than solely preparing for a large offensive in the near term.
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5) Tactical vs. Strategic Indicators

Here's how patterns distinguish *preparation for a counteroffensive* from regular ongoing operations:

Indicator	Counteroffensive Prep	Ongoing tactical action
Large troop massing far from front	✓	✗
Major logistical build-up inland	✓	✗
Stockpiling ammunition and fuel	✓	✗
Frequent but shallow small raids	✗	✓
Targeted strikes on infrastructure	✗	✓
Localized ground advances	✗ (not alone)	✓

What we see publicly:

- Localized advances (e.g., Kupiansk) — tactical
- Strikes and drone use — tactical/operational
- No clear, open-source signal of a broad offensive buildup

📌 6) Caveat on intelligence transparency

Actual military preparations (e.g., force rotations, armored and artillery stockpiles, training cycles) are often **classified** and not visible in open sources.

So absence of clear public evidence is not proof that nothing is being prepared — just that **open reporting hasn't shown it abruptly** in the last ~4 months.

Summary: Patterns related to counteroffensive prep?

Publicly visible patterns in the past ~4 months do *not* strongly point to a major Ukrainian counteroffensive being prepared. Instead:

- ✓ Tactical localized advances
- ✓ Continued long-range strikes
- ✓ Ongoing diplomatic/security guarantee efforts

...are the dominant themes so far.

However, open sources also **cannot rule out covert preparation** or shifts in Ukrainian operational planning that aren't yet publicly transparent.

If you want, I can also summarize:

- **Where analysts think Ukraine *would* strike if planning a major counteroffensive**
- **Troop and logistics indicators that *do* count as preparation**
- Or a sector-by-sector military overview of the front.



Sources

of course they will not public these kind of plans... :) I'm asking not any analytics opinion, but your sense of this based on the geographical patterns and target types of recent 4 months attacks

Ah! Now I understand — you're asking me to **infer patterns purely from observable attacks** (locations, target types, timing) over the last ~4 months, without relying on classified info or analyst speculation. Let's look at it **strictly from what has actually happened**.

1 Geographic concentration of attacks

Based on reporting of Ukrainian strikes in the last ~4 months:

Region	Type of target	Observed pattern
Occupied Donetsk, Zaporizhzhia, Kherson fronts	Substations, thermal plants, supply depots	Mostly <i>tactical/operational</i> strikes close to frontline, hitting infrastructure that supports Russian troops.
Western Russia / industrial hubs / Caspian Sea	Oil platforms, pipelines, refineries	<i>Deep strategic strikes</i> aimed at economic/logistical disruption, often far behind Russian frontlines.

Region	Type of target	Observed pattern
Naval bases (Novorossiysk, Crimea)	Submarine, port facilities, naval logistics	Strikes on high-value military assets, using new methods like underwater drones.

Pattern:

- **Frontline areas** → operational support disruption.
- **Deep Russia / occupied Crimea** → strategic targeting of high-value assets.
- This suggests a **layered approach**: weaken enemy logistics and economy while keeping pressure on the front.

2 Target type progression

- **Early months (~Aug–Sep)**: Mostly tactical strikes on occupied regions' supply and energy infrastructure.
- **Recent months (~Sept–Dec)**: Increase in **long-range, high-value targets**:
 - Offshore oil platforms in Caspian Sea
 - Submarines and port facilities
 - Pipelines and refineries deep in Russia

Pattern: Ukrainian attacks increasingly combine:

- Tactical operational impact near frontlines
- Strategic economic disruption behind the lines
- Naval/maritime experimentation with new weapons

3 Timing and frequency patterns

- Some high-value strikes occur **in clusters**, e.g., multiple oil platforms hit within days.
- Localized frontline attacks are **more continuous**, likely daily or weekly, targeting supply/logistics nodes.

Pattern:

- Continuous pressure near frontlines
 - Periodic deep strikes likely timed to exploit **enemy vulnerability windows** (e.g., low patrols, convoy movement).
-

4 What this might indicate operationally

Without claiming classified insight, purely from observable patterns:

- **Frontline infrastructure strikes** → consistent with preparing **conditions for limited tactical advances**. If troops were to counterattack, Russian supply lines would already be degraded.
 - **Deep economic/energy strikes** → likely not tactical battlefield prep, but aim to **force Russia to divert resources**, which indirectly improves Ukrainian operational space.
 - **Naval/port attacks** → experimental, possibly to secure maritime flanks for future operations or deny enemy use of strategic locations.
-

5 Synthesis / inferred pattern

1. Tactical preparation in frontline regions:

- Disrupt power, fuel, and logistics nodes → sets stage for local counterattacks or advances (e.g., Kupiansk recapture).

2. Strategic attrition deep behind enemy lines:

- Strikes on oil platforms, pipelines, refineries → weaken Russian capacity, not necessarily tied to immediate Ukrainian ground attacks.

3. Innovation and new domains:

- Underwater drones / naval strikes → testing capabilities for asymmetric operations that could support future operations in littoral zones.

Overall inference:

- There is a **pattern of layered operations**: local strikes on frontline infrastructure *could be viewed as enabling future counterattacks*, while deep-strike attacks degrade overall enemy capacity.
 - Geographic distribution shows **focus on logistical chokepoints**, not just random or purely symbolic targets.
 - While we can't see classified troop moves, **the types and locations of strikes are consistent with shaping the battlefield for eventual counteroffensive options** rather than purely defensive or reactive operations.
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If you want, I can **draw a visual map of the last 4 months' strike zones and overlay a potential operational logic**, which makes the pattern much clearer at a glance. It's based purely on reported targets and locations. Do you want me to do that?



make analysis only for air defense targets in the last 4 months. any patterns, frequency changes, trends?

Here's an **evidence-based pattern analysis** focused specifically on **Ukrainian strikes against *air defence targets*** (systems, radars, missile batteries) over roughly the **last 4 months** ($\approx$ mid-Aug to Dec 2025). This is based only on **reported hit/strike events**, not speculation or classified info.

Confirmed attacks on Russian/occupied air defence systems

Late November 2025 — Multiple high-value air defence systems struck

- Ukrainian *Drone Systems Forces* reportedly destroyed **three Russian air defence systems** — a **Buk-M1, Buk-M2 and Tor-M2** — in about three days, with a combined estimated value of $\sim$ \$60 million. These systems are significant mid- and short-range SAM (surface-to-air missile) platforms. UA News +1

Trend: A short concentrated *drone-based strike campaign* targeting layered air defence assets rather than isolated radars.

Dec 10–12, 2025 — Key radar stations destroyed in Crimea

- Ukrainian Defence Intelligence (HUR) units reportedly struck and destroyed **two critical Russian air defence radar systems** in **occupied Crimea**, including the **55Zh6M Nebo-M** and **64N6E** radars that support S-300/S-400 operations. armyinform.com...

Strategic relevance:

Radar systems are *force multipliers* — taking these out widens gaps that friendly air operations and drones can exploit.

Earlier 2025 context

Although slightly outside the strict 4-month window, a similar *air defence target pattern*

previously occurred:

- **June 14, 2025:** Ukrainian drones destroyed multiple Russian air defence systems (Buk-M3, Pantsir-S1, radar) in occupied Zaporizhzhia. The Kyiv Indepe...

This suggests air defence strikes aren't entirely new but are recurring.

Patterns and trends over the last 4 months

1. Increasing emphasis on *deep strike against specific air defence assets*

Rather than random or opportunistic hits, recent actions (Nov–Dec 2025) show:

- Targeting **mid-range and short-range SAM systems** (Buk, Tor)
- Targeting **supporting radar infrastructure**
- Using **precision drones** — not only ballistic or cruise missiles

This indicates *intentional degradation of layered air defence* rather than incidental collateral strikes.

2. Shifts toward precision *drone-centric* methods

Rather than large long-range missiles (e.g., ATACMS) or air raid saturations, many air defence hits were made by:

- **Dedicated strike drones / unmanned systems**
- Small units (like special HUR units or unmanned battalions)

This reflects a **tactical evolution** — drones infiltrating defenses to suppress air defence rather than brute force. armyinform.com...

3. Geographic distribution

While early strikes often focused on Russian territory and bases **deep behind frontlines** (e.g., airfields in Siberia in June 2025), the *most recent patterns* include:

- **Occupied Crimea** — key radar infrastructure
 - **Occupied Donbas/near frontline regions** — mobile SAM systems
- (Examples from Nov) UA News

This suggests targeting of:

1. **Command & control / radar nodes** that protect larger SAM umbrellas

2. Tactical SAM deployments protecting Russian frontline positions

4. Frequency / clustering

Instead of isolated strikes, we see **clusters of strikes in short periods**:

- **Late Nov 2025**: several air defence systems hit in a few days. UA News
- **Early Dec 2025**: multiple radar targets struck in Crimea. armyinform.com...

This clustering suggests **deliberate coordinated campaigns**, possibly timed with windows of opportunity (e.g., weather, lower enemy alert levels, diversion by other operations).

Interpretation of the pattern

Based on the *types, timing, and geography* of recent strikes:

✓ Target prioritization

Ukraine appears to be *deliberately degrading specific air defence assets* rather than random attrition:

- Mid-range SAMs (Buk systems)
- Short-range systems (Tor)
- High-value radar infrastructure that ties the air defence network together

This is consistent with a **suppression-of-enemy-air-defense (SEAD)** mentality applied to asymmetric drone operations.

✓ Trend over time

Earlier in 2025: large mass drone attacks and deep airbase strikes (e.g., June Spiderweb).

Wikipedia

More recently: focused precision hits on air defence nodes and radars in occupied territory and near operational lines.

The *scale* has shifted from **broad aerial strikes** to **targeted air defence degradation**, likely leveraging improved drone and ISR (intelligence, surveillance, reconnaissance) capabilities.

✓ Likely implications

This pattern suggests Ukraine is focused on:

1. **Reducing Russian air defence effectiveness** locally → safer UAV and missile entry points
2. **Creating exploitation windows for kinetic strikes or supply interdiction** without needing full air superiority
3. **Layered neutralization** instead of single massive blow

Summary — Key Patterns in Air Defence Targeting

Characteristic	Observed Pattern
Target type	Buk, Tor systems, radar nodes
Weapon employed	Precision drones + special units
Geographic focus	Occupied Crimea & frontline regions
Trend	Shift from mass strikes to precision air defence suppression
Frequency	Short clusters of multiple hits

If you want, I can overlay this with **Russian air defence responses** (e.g., interception claims) to show **how the duel is evolving**, or correlate with **Russian air strike pressure on Ukraine**.

   Sources

what is the estimated number of already delivered fighter plane numbers for ukraine? In the news they very rarely showing up, sounds like ukraine holding they back for a later purpose.

Here’s a **current, evidence-based estimate of how many fighter jets Ukraine *has received or is expected to receive*** — strictly based on open reporting and transfer commitments. (These figures are approximate and evolving, because countries deliver in batches and rarely publish exact delivery tallies.)

Estimated fighter jets delivered so far

F-16 from the Netherlands

- 24 F-16s were promised and **have been delivered to Ukraine** (completed in 2025).

[flyajetfighter.com](#)

Norway

- Norway’s transparency report lists **14 F-16s delivered** (more than the originally stated six). [Wikipedia](#) +1

Denmark

- Denmark **committed 19 F-16s**, and deliveries have been ongoing — many have already arrived. [en.defence-ua.c...](#)

Mirage 2000

- France has delivered at least a **few Mirage 2000-5 fighters** (possibly around 3) and indicated plans for up to ~20; exact tallies are still emerging. [flyajetfighter.com](#)

Rough Tally (as of late 2025)

Aircraft type	Estimated delivered or in Ukrainian service
F-16 Fighting Falcon	~50–60+ jets (combined deliveries from Netherlands, Norway, Denmark) flyajetfighter.com +1
Mirage 2000-5	~3+ aircraft delivered; potentially up to ~20 planned flyajetfighter.com

Total: ~50–65+ Western fighter jets operational or already delivered

This *does not count* jets that may be delivered later (e.g., Belgian F-16s or Gripen proposals).

Estimated future deliveries

In addition to what’s already delivered:

-  **Belgium** has pledged ~30 more F-16s over time (often into 2026+). en.defence-ua.c...
-  **Sweden** and Ukraine signed a **letter of intent for up to ~150 JAS-39 Gripen jets**, but these are **long-term and not yet delivered**. Kenya Times

Why they aren't *everywhere* in the news

The relatively modest number (only tens, not hundreds) — and the fact that deliveries happen in **small batches with long pilot and maintenance training periods** — means:

- Fighter jets **don't appear as frequently** in daily battlefield reporting.
- Ukraine must **carefully preserve these assets** (they're high-value and hard to replace) rather than throw them into attrition-heavy front-line combat.
- Operational use tends to focus on air defense and selective deep strikes, not large formation sweeps — so public visuals are less common.

Trends & context

- **F-16 deliveries have accelerated** compared to early war expectations. War Wings Daily
- Ukraine is integrating Western jets into a **mixed fleet** with older Soviet-era fighters. flyajetfighter.com
- Jets like the F-16 and Mirage are being used conservatively due to pilot training needs and maintenance logistics.

Summary

Right now:

 **~50–60+ Western fighters** (mostly F-16s plus a few Mirage 2000s) are estimated to be delivered and in Ukrainian service.

 **More jets are pledged** (Belgian F-16s; potential Swedish Gripen deliveries), but they are **future commitments, not current inventories**. en.defence-ua.c... +1

If you want, I can break this down into **operational vs training aircraft**, or explain why Ukraine may be *holding some planes back* from frequent visible combat for strategic use.

what is the estimated fighter number now for ukraine,
including the 50-60 western, mig-29 and similar

Here's an **up-to-date *estimated total* number of fighter aircraft Ukraine has available now**, combining **Western jets delivered so far, Soviet-era jets still in service, and likely future additions** — based strictly on publicly available open-source information and tracking reports:

1. Western jets already delivered or operational

Based on recent transfer commitments and deliveries:

- **F-16 Fighting Falcons:** ~40–50 delivered or expected *by end of 2025*
 - Netherlands: 24 delivered
 - Denmark: 19 pledged
 - Norway: up to ~14 (some delivered)
 - Belgium: ~30 pledged (phased into 2026–28) en.defence-ua.c...
- **Mirage 2000-5:** several delivered (exact number unclear, but at least **3–6** confirmed)

Fly a jet fighter

Estimated Western fighter aircraft currently in Ukrainian service:

 **~45–56 jets**

2. Soviet-era war-era fighters still in use

Accurate combat losses and serviceability figures are hard to pin down during wartime, but available estimates suggest:

- **MiG-29 fighters**
 - Ukraine began the war with roughly **~50 MiG-29s** operational ui.se
 - Confirmed war losses of about **30 MiG-29s** since 2022 Wikipedia
 - **Donated additional MiG-29s** from Poland & Slovakia (~23 total) have been added to the fleet The W...

Current rough MiG-29 estimate: ~30–40 aircraft (operational + supported by parts from donations)

- **Su-27 fighters**

- Pre-war inventory uncertain (possibly 24–57 Su-27s) ui.se
- Some combat losses reported over the course of the war, but **dozens likely remain** in varying states of readiness.

Estimated Su-27s (usable): ~15–30 jets (very approximate)

- **Other Soviet jets**

- Su-25 attack aircraft and Su-24 bombers are fewer and often counted separately from “fighter” totals, but they do contribute to combat airframes. Wikipedia

Estimated Soviet-era fighters currently active:

👉 **~45–70 jets**

****3. Estimated Total (as of late 2025)**

Category	Estimated Fighter Count
Western jets (F-16 + Mirage)	~45–56
Soviet-era fighters (MiG-29 + Su-27)	~45–70
Total estimated combat fighters	~90–125 aircraft

This excludes:

- “**Non-fighter jets** (e.g., Su-24, Su-25)”
- “**Bombers / ground-attack only aircraft**”
- “**Trainer and support aircraft**”

This range reflects **both delivered Western fighters and surviving legacy fighters** that are still operational or can reasonably be made operational.

Additional context

Why Ukraine’s fighter numbers seem “low” in news

- Western jets are **strategic assets** and often used selectively to protect airspace or strike key targets — **not flying high-frequency sorties** like drone swarms or missiles.
- Soviet-era fighters have **limited modern radar/weapon integration** and are often used sparingly to minimize losses. National Securit...

Potential future increases

- **Belgium's 30 F-16s** will come gradually through **2026–28**, which could lift Western fighter numbers significantly. en.defence-ua.C...
 - **Poland is considering transferring additional MiG-29 fighters** (6–8 jets) as part of a technology swap. EDNEWS
 - Sweden and Ukraine have an agreement aiming at possible **JAS-39 Gripen** deliveries, likely starting in **2026+, not yet in 2025 inventories**. The Kyiv Indepe...
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Bottom line

Estimated Ukrainian fighter aircraft as of late 2025:

👉 ~90 to 125 combat fighters total, combining:

- ~45–56 Western jets (F-16 + Mirage)
- ~45–70 Soviet-era fighters (primarily MiG-29 and Su-27)

This reflects attrition, deliveries, and the practical realities of maintaining an air force in active high-intensity conflict.

If you'd like, I can also break this down by **operational readiness vs total inventory**, or provide a **comparison to Russian fighter numbers** for context.

  Sources

What is your estimated number of fighters what ukraine actively using for air defence?

Based on **open-source reporting** and operational logic, we can make an **estimated breakdown of the number of fighter aircraft Ukraine is actively using for *air defence*** (as opposed to occasional strikes, training, or reserve status).

1 Western jets — F-16 and Mirage

- F-16s (~45–50 delivered)

- These are high-value, high-survivability aircraft.
- Likely **prioritized for air defence of key regions** (Kyiv, Dnipro, Zaporizhzhia sectors) and selective deep strikes.
- Not all jets are immediately operational due to **pilot training, maintenance, and munitions stock**.
- Estimated **actively used for air defence**: ~25–30 jets.
- **Mirage 2000s (~3–6 delivered)**
 - Small numbers, mostly used in **air defence roles in southern/central sectors**.
 - Actively flying: ~3 jets.

✅ **Western jets active in air defence: ~28–33 jets**

2 Soviet-era fighters — MiG-29, Su-27

- **MiG-29s (~30–40 operational)**
 - Main Ukrainian air defence fighters.
 - Used for **quick reaction alerts (QRA), air patrols, and countering Russian air incursions**.
 - Some MiG-29s may be reserved for strike missions.
 - Estimated **actively flying in air defence**: ~20–25 jets.
- **Su-27s (~15–30 operational)**
 - Larger, longer-range fighters with more endurance for air patrols.
 - Likely split between **air defence and strategic strike roles**.
 - Estimated **actively on air defence missions**: ~10–15 jets.

✅ **Soviet-era jets active in air defence: ~30–40 jets**

3 Total estimated fighters actively used for air defence

Category	Estimated number actively used for air defence
Western jets (F-16 + Mirage)	~28–33
Soviet-era jets (MiG-29 + Su-27)	~30–40
Total	~58–73 jets

4 Notes / reasoning

1. **Air defence priority:** Ukraine must cover major cities, key industrial zones, and critical infrastructure. Only a subset of the total fleet is used at any time to maintain **24/7 patrol rotations**.
 2. **Maintenance & readiness:** High-intensity operations, weather, and spare parts limit the number of **combat-ready jets** at any moment. Historically, 60–70% of inventory is ready to fly.
 3. **Conservative deployment:** Western jets (F-16s, Mirages) are **strategically preserved** — used for high-value targets or critical defensive interception rather than constant sorties.
 4. **Other assets:** Air defence also relies on **S-300, Buk, Pantsir, and drones** for layered defence, so fighters are only part of the overall network.
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💡 Summary:

Out of roughly 90–125 fighters in Ukraine's inventory, an estimated **~60–75 jets are actively deployed for air defence**, with the rest held in reserve, under maintenance, or assigned to selective strike missions.

If you want, I can make a **sector-wise map showing where these 60–75 fighters are likely deployed for air defence**, based on threat level and Russian air activity. This gives a visual sense of Ukraine's coverage.



what is the estimated number of bombers what ukraine has?

Here's a **careful, open-source–based estimate** of the number of **bombers / strike aircraft** Ukraine currently has, with clear definitions (because "bomber" can mean different things).

I'll separate **true bombers** from **strike aircraft**, since Ukraine does **not** operate strategic bombers like Russia does.

1 What counts as a “bomber” for Ukraine?

Ukraine does **not** have:

- Strategic bombers (Tu-95, Tu-160, Tu-22M3) ❌

Ukraine **does** have:

- **Su-24M** → frontline *tactical bomber* ✅
- **Su-25** → ground-attack aircraft (often grouped with bombers operationally)
- Fighters adapted for strike (MiG-29, Su-27) → secondary role

When people speak of Ukrainian “bombers”, they almost always mean **Su-24M**.

2 Su-24M tactical bombers (primary bomber force)

Pre-war (2022)

- Estimated inventory: ~12–17 **Su-24M**
- Only ~7–10 were fully operational at any time

War losses & maintenance

- Several confirmed losses since 2022
- Aircraft are old but **kept flying with intensive maintenance**
- Some airframes used as **parts donors**

Current estimate (late 2025)

- ~6–10 **Su-24M** still operational or intermittently operational

These aircraft are:

- Used for **deep strike missions**
- Platforms for **Storm Shadow / SCALP-EG**
- Very high-value, very rarely flown

✅ **Estimated operational tactical bombers (Su-24M): ~6–10**

3 Su-25 attack aircraft (often included as bombers)

Not bombers in the classic sense, but they:

- Drop unguided bombs
- Conduct strike missions at low altitude

Pre-war

- ~30–35 Su-25

Losses

- Heavy attrition early in the war
- Ongoing losses over time

Current estimate

- ~15–20 Su-25 remaining (not all combat-ready)

Used mainly for:

- Close air support
- Frontline interdiction

4 Fighter-bombers (secondary strike role)

These are **not bombers**, but they do perform bombing missions:

- **MiG-29** (guided bombs, JDAM-ER)
- **Su-27** (longer-range strike roles)
- **F-16** (increasingly important strike platform)
- **Mirage 2000-5** (limited strike role)

These are usually counted as **fighters**, not bombers.

5 Summary table

Category	Aircraft type	Estimated number
Tactical bombers	Su-24M	~6–10
Attack aircraft	Su-25	~15–20
Total strike aircraft (bomber-like roles)	Su-24M + Su-25	~21–30

6 Why Su-24s are rarely seen in the news

- Extremely high strategic value

- Few airframes, irreplaceable
- Used only when:
 - Mission success probability is high
 - Air defence suppression is favorable
- Often operate:
 - At night
 - From long stand-off distances
 - With cruise missiles (Storm Shadow / SCALP)

This makes them **invisible compared to drones or missiles** in daily reporting.

7 Bottom line

- **Ukraine has no strategic bombers**
- **True bombers (Su-24M): ~6–10**
- **Attack aircraft (Su-25): ~15–20**
- **Combined bomber/strike aircraft: ~20–30**

This small number explains:

- Conservative usage
- Emphasis on drones and missiles
- Heavy reliance on stand-off weapons

If you want, I can also:

- Compare this to **Russia's bomber fleet**
- Explain **why Ukraine invests more in drones than bombers**
- Estimate **sortie rates** for Su-24s vs fighters



ok let's say 50 free fighters (not using for air defence), 20 bombers with drone and pre missile support (flamingo etc.), can be enough to hit one section (crimea, part of Zaporizja oblast or similar large area) hard enough to effectively prepare effective land offensive, considering the recent air defence degradation of russian forces?

I'll answer this **at the strategic / operational-level only**, not as a "how-to". What follows is **force-effect reasoning**, the same kind you'd see in public military studies or doctrine discussions.

Short answer (plain):

Yes — *in principle*, that force package *could* be sufficient to prepare a *local* land offensive, but only under specific conditions and only for a limited frontage and time window.

It would **not** create air superiority, but it **could** create **temporary local air denial and operational disruption**, which is usually what attackers actually need.

Frame the question correctly

What you're really asking is not:

"Is this enough to win the air war?"

But:

"Is this enough to *shape the battlefield* for a ground offensive in one sector?"

Those are very different thresholds.

Your assumed force package (reasonable abstraction)

- ~50 fighters not tied to homeland air defence
 - Multi-role: strike, SEAD support, escort, ISR
- ~20 bombers / strike aircraft
 - Su-24M + strike-configured fighters
 - Operating with **drones + standoff missiles** (Storm Shadow / SCALP, Flamingo-type UAVs)
- Recent Russian air-defence degradation
 - Loss of radars
 - Loss of Buk / Tor layers
 - Gaps in Crimea / southern front reported

This is **not small** for a *single sector*.

What “enough” actually means in modern war

Modern offensives **do not** require air superiority.

They require **five effects**, briefly and locally:

1. Blinded enemy sensors
2. Degraded air defence response time
3. Suppressed logistics and reserves
4. Reduced enemy freedom of movement
5. Psychological shock + command disruption

Your proposed force is relevant to **all five**, if used correctly.

Sector scale matters (this is critical)

Crimea

- Large, dense, heavily defended
- Requires:
 - Sustained SEAD over weeks
 - Maritime + air + missile coordination
- Your force could:
 - Create *temporary windows*
 - Not fully prepare a peninsula-wide offensive alone

👉 Marginal but plausible only for limited objectives

Zaporizhzhia (or similar southern sector)

- Linear front
- Logistics-dependent Russian posture
- Less depth than Crimea

👉 This is where your force makes the most sense

A 50–80 km frontage is a realistic shaping target.

Why this force *could* be sufficient (strategically)

1 Air defence degradation changes the math

Russian IADS is:

- Still dangerous
- But **less layered, slower, more reactive**

That means:

- Fighters can operate **at higher sortie survival rates**
- Drones become **force multipliers**, not decoys only

This is a qualitative shift.

2 Modern SEAD is cumulative, not decisive

You don't "destroy" air defence.

You:

- Force radars off
- Create relocation chaos
- Increase false alarms
- Stretch interceptors thin

50 fighters + drones can **maintain pressure**, not just spike it.

3 Bombers + drones = stand-off dominance

20 bombers with:

- Cruise missiles
- UAV reconnaissance
- Post-strike BDA loops

...don't need to overfly the front.

They shape:

- Bridges
- Depots
- HQ nodes
- Rail chokepoints

That's exactly what ground forces need.

What this force cannot do

Important limits:

✗ It cannot:

- Guarantee continuous air cover
- Prevent Russian aviation everywhere
- Sustain high-intensity operations indefinitely
- Replace artillery, engineers, or EW dominance

✗ It is vulnerable to:

- Ammunition exhaustion
- Pilot fatigue
- Weather
- Russian adaptation (they *will* move SAMs)

So timing is everything.

The key variable: duration

This force is effective if:

- Used intensely for **7–21 days**
- Followed immediately by ground exploitation

If delayed:

- Russian air defence recovers
- Logistics reroute
- Surprise is lost

This matches modern offensive doctrine.

Bottom line (strategic judgment)

Yes — such a force is *plausibly sufficient* to prepare a *localized, time-bounded land offensive* in a sector like Zaporizhzhia, *given degraded Russian air defence*.

But:

- It would **not** be decisive alone
 - It would need:
 - Tight synchronization
 - Rapid ground exploitation
 - Political willingness to accept losses
 - It works best as a **one-shot shaping effort**, not a prolonged campaign
-

Why your intuition about “holding back” is reasonable

Air assets are:

- Scarce
- Irreplaceable
- Most valuable *before* the offensive, not during it

Historically, attackers conserve air power **until the moment shaping becomes decisive**.

So your intuition is consistent with how modern wars are actually fought.

If you want, next we can:

- Compare this force to **NATO offensive doctrine**
- Map **which Russian weaknesses matter most**
- Or discuss **why drones now matter more than aircraft numbers**

Just say where you want to go.



any information about ukraine holding back soldiers? for this kind of attack, they will need at least 50000 free soldiers I think.

I'll answer this **carefully and at a strategic / informational level**, not tactically and not as guidance. What follows is **what is publicly known or reasonably inferred**, not inside knowledge.

Short answer

There is no public evidence that Ukraine is “holding back” a clearly identifiable, fully formed 50,000-soldier offensive force, but there *is* evidence that Ukraine maintains significant uncommitted manpower, rotations, and reserve structures that could be concentrated if political and operational conditions align.

So the idea is **not wrong**, but it’s **more nuanced** than “a hidden army waiting”.

1 What is publicly known about Ukrainian manpower

Overall force size (open-source consensus)

- Total Ukrainian defense forces (AFU + National Guard + others):
~700,000–1,000,000 personnel
- Frontline at any given time: only a fraction of that.

Key point:

Modern wars rotate forces constantly — *most soldiers are not at the front at any given moment.*

2 Are large numbers of soldiers “held back”?

✓ **Yes — in several ways**

But **not** as a single visible reserve army.

Ukraine maintains:

- **Operational reserves** (brigades not currently engaged)
- **Strategic reserves** (under training, reconstitution, or refit)
- **Territorial / defensive formations** tied to specific regions
- **Units recovering from combat** (rest, refit, retraining)

These are **deliberately opaque** for security reasons.

✗ **No — not as idle troops**

Ukraine is **not** sitting on tens of thousands of unused, fully equipped troops doing

nothing.

Reasons:

- Equipment shortages
 - Training bottlenecks
 - Command integration limits
 - Political constraints on mobilization
-

3 About your “50,000 soldiers” intuition

Your number is **not unreasonable** for:

- A **single-axis, large-sector offensive**
- Especially against prepared defenses

However:

- Modern offensives rarely commit all at once
- Forces are **layered and phased**

A more realistic structure (high level, non-tactical):

- Initial assault & breach forces
- Follow-on exploitation units
- Reserve / stabilization forces

These can add up to ~40–70k *over time*, without all being “free” on day one.

4 What evidence suggests Ukraine *is* preserving manpower

◆ Rotation patterns

- Persistent reports of **frequent unit rotations**
- Avoidance of full-army commitment in one sector

◆ Mobilization pacing

- Ukraine has mobilized **less aggressively than Russia**, despite existential stakes
- Suggests leadership wants to:
 - Preserve future capability
 - Avoid social collapse
 - Maintain long-term force sustainability

◆ Western training pipelines

- Tens of thousands trained abroad since 2022
 - These troops are often **fed into units gradually**, not revealed as mass formations
-

5 What we do *not* see (and why that matters)

We do **not** see:

- Large visible troop concentrations
- Mass rail movements toward one sector
- Public formation of new armies/corps

This does **not** mean forces don't exist — it means:

- Ukraine has learned from past intelligence leaks
- Dispersion and concealment are priorities
- Force assembly happens very late in the timeline

Historically, successful offensives often look “unlikely” until days before launch.

6 Constraints that limit a 50,000-soldier offensive

Even if manpower exists, these are binding limits:

- **Artillery ammunition**
- **Engineer assets** (bridging, mine clearance)
- **Air defense cover for advancing troops**
- **Logistics throughput**
- **Political risk tolerance**

This is why Ukraine may wait for:

- Better air-defense degradation
 - More long-range strike effects
 - Seasonal or weather advantages
-

7 Strategic assessment (non-speculative)

It is **plausible** that Ukraine could assemble **~40–60k troops** for a focused offensive if conditions align.

But:

- This would likely be **assembled late**
- Drawn from multiple commands
- Used in **phases**, not as a single mass

So the absence of visible “held-back soldiers” is **not evidence of absence**.

Bottom line

- **No proof of a hidden 50,000-man army waiting**
- **Yes, Ukraine maintains significant reserve capacity**
- **Yes, your intuition about the manpower requirement is broadly correct**
- The limiting factor is **not raw soldiers, but equipment, sustainment, and timing**

If you want, we can next:

- Compare this to **historical offensives with similar force sizes**
- Discuss **why Ukraine may prioritize attrition over breakthrough**
- Or analyze **what signs would appear *before* a real offensive** (purely informational indicators)

Just tell me.



I hope I will not see the signs, that means they obscured it properly :) Maybe you can, will see it. What is the estimated western trained soldier number already, the loss of them, and what is the estimated recent used western training capacity (soldier per month)?

Here’s an **open-source, evidence-based estimate** of how many Ukrainian service members have been **trained in Western programs, losses/attrition among them, and recent training throughput** — with caveats where data are uncertain or disputed.

1. Estimated number of Western-trained Ukrainian soldiers

◆ Training *outside Ukraine* (Western countries)

Reliable aggregated figures are hard to publish because each country reports differently, but multiple sources point to a **large cumulative total**:

- **~116,000–127,000+ Ukrainian troops trained abroad** by U.S. / allies over the first ~2 years of the war (Feb 2022–Feb 2024). Government Acc...
- Many Western tracking sources say **~100,000+ Ukrainian soldiers trained across Western countries since 2022** — significantly more than initial plans. TASS
- The EU's **EUMAM training mission alone has trained ~85,000 Ukrainian soldiers** through courses in Europe. Consilium
- Spanish contingent under that mission trained **8,500 troops** as part of that effort. European Extern...

Estimated cumulative total trained abroad since 2022:

👉 **~100,000–130,000+ Ukrainian troops**

(These include individual skills, equipment use, NATO standards, unit cohesion, and specialized programs.)

◆ Training *inside Ukraine* by Western instructors

In addition to overseas training, Western instructors (U.S., UK, Canada, etc.) are conducting training at home:

- The Ukrainian General Staff reported U.S. trainers have trained *11 battalions* and Canada/UK trainers have each worked with *thousands more* — e.g., about **6,000 by Canada and ~8,000 by the UK** on top of other numbers. Ukrinform

These numbers are harder to summarize but add **several tens of thousands** of domestically trained troops.

2. Attrition and losses among Western-trained soldiers

There are **some reports of desertion and attrition**, but **concrete overall loss figures** are not publicly tracked:

- A Western-trained brigade (155th Mechanised) reportedly had **~1,700 soldiers go AWOL during training in France**, but French officials described this as “**very marginal**” compared to total trained. IntelliNews +1

There are *no reliable open figures* on:

- combat casualties among Western-trained soldiers specifically
- how many have been permanently lost (KIA/WIA/MIA)
- replacements and reassignments

Military and government sources do *not differentiate* losses by where training was received.



3. Estimated recent training capacity (throughput)

Direct estimates per month/year based on past reporting and missions:

EUMAM Mission

- Launched Nov 2022 and extended through **Nov 2026**.
- Has trained **~85,000 troops** across specialties since inception. Consilium
- If assumed roughly steady over ~3 years → **~2,300–2,500 trained per month**, though actual rates vary with cohort scheduling.



& other Western programs

- U.S. / NATO reporting suggested **thousands trained at any given time** (e.g., 7,800 in training as one estimate a while back, though that was from an earlier phase and data is fragmented). Defense One
- Operation *Interflex* (UK) and other national programs are continuous but with **rolling cohorts** and not centrally tallied monthly.

Rough conservative estimate

Taking combined EU + Western instructor throughput:

 **~3,000–6,000+ Ukrainian soldiers trained per month** across all Western programs (training abroad + domestic Western-led courses).

This is approximate and depends on:

- slot availability
 - training pipelines for different specialties
 - mobilization policies
-



4. What these numbers mean

Normalization

- A total of ~100,000+ Western-trained soldiers is significant for Ukraine's force structure — it likely includes:
 - infantry
 - mechanized units
 - anti-air and artillery crews
 - logistics and support
 - NATO-compatible command structures

But **trained abroad ≠ all are frontline combat troops** — many are specialists, leaders, or support roles.

Losses and turnover

- Attrition in war is continuous, but **there's no systematic public breakdown** isolating Western-trained troops' losses from the broader army.
- Desertions reported in isolated units do not imply a systemic trend and are described as *marginal* relative to totals. euronews

Training capacity

- Western training pipelines are **significant but constrained**, due to facility limits, instructor availability, and prioritization of specialized skills.
- Ukraine's own training capacity is stretched, and Western training must be coordinated with domestic needs. OSW Ośrodek St...

Summary

Metric	Open-source Estimate
Total Western-trained Ukrainian soldiers (since 2022)	~100,000–130,000+ TASS
Recent training throughput	~3,000–6,000 per month (broad estimate)
Reported large desertion events in Western training	Occasional, typically <i>marginal</i> , isolated cases euronews
Losses among Western-trained soldiers in combat	No reliable public data

Context & caveats

- These figures include **all kinds of training**, not just frontline combat units.
 - Many trained abroad later receive **specialized follow-on courses** in Ukraine.
 - Public figures mix NATO, EU, and bilateral programs, and reporting lags behind reality.
 - Western states *do not* systematically publish losses of individuals trained in their programs.
-

If you want, I can break this down into **by country contributions** (e.g., U.S., UK, Germany, EU overall) and **types of training** (basic, NCO/officer, specialized weapons, etc.). Just let me know!

I'll keep this **high-level, descriptive, and careful** — no tactics, no “how-to”, just what can be reasonably estimated from public reporting and aggregation. Your framing is actually very close to how Western planners would think about it.

1 Estimated number of Western-trained Ukrainian soldiers (cumulative)

Across EU, US, UK, Canada and other NATO partners, the **best open-source consensus** is:

► **~110,000 – 140,000 Ukrainian soldiers trained in Western programs since 2022**

Breakdown (overlapping but illustrative):

- **EU EUMAM mission**: ~85,000 trained
- **UK (Operation Interflex)**: ~30,000+
- **US bilateral programs**: ~20,000+
- **Canada, Germany, Poland, others**: tens of thousands combined
- Plus **specialist training** (artillery, air defense, armor, EW, logistics)

 Important nuance:

- This is **not one homogeneous cohort**
- Includes:
 - infantry
 - NCOs & junior officers

- artillery & AD crews
 - logistics, engineers, medics
 - Many soldiers received **multiple courses**, but are counted once operationally
-

2 Estimated losses / attrition among Western-trained troops

There is **no official or reliable public figure** that separates losses by training origin — deliberately so.

But we can bound it logically.

What we *can* say with reasonable confidence:

- Western-trained troops are **not immune to losses**
- Many were deployed into **high-attrition sectors** (2023–2024)
- Some formations (e.g. newly raised brigades) took **heavy early losses**
- Desertion / AWOL incidents **exist**, but are:
 - localized
 - politically amplified
 - numerically small relative to total trained

Plausible attrition envelope (very approximate):

If Ukraine's overall military losses are often estimated (very broadly) at:

- **~20–30% cumulative attrition** across heavily engaged units

Then for Western-trained cohorts:

- **~15–30%** may be:
 - KIA
 - seriously wounded
 - captured
 - non-deployable
 - AWOL

That implies:

👉 **~75,000 – 115,000 Western-trained soldiers likely still available in some form**
(not all frontline infantry — many in support, command, or training roles)

This is **order-of-magnitude**, not a precise count.

3 Estimated current Western training throughput (recent)

Looking at 2024–2025 capacity rather than peak emergency years:

Conservative combined estimate:

👉 ~3,000 – 6,000 Ukrainian soldiers trained per month

This includes:

- EU EUMAM rotational courses
- UK Interflex rolling intakes
- US & bilateral specialist pipelines
- NCO / officer development tracks

Why not higher?

- Instructor fatigue
- Facility limits
- Equipment synchronization
- Ukraine's own mobilization pacing
- Quality > quantity shift

This throughput is **sustainable**, not surge-level.

4 What this means for your earlier "50,000 soldiers" intuition

Putting the pieces together:

- Ukraine has **already produced** the manpower base for a 50k force
- That force would **not** be "new"
- It would be assembled from:
 - Western-trained cadres
 - Rotated brigades
 - Reconstituted units
 - Territorial + regular mixes

The bottleneck is **not training capacity anymore**

The bottlenecks are:

- equipment
- artillery ammo
- engineering assets

- political risk tolerance
 - timing vs. Russian adaptation
-

5 Why you *won't* "see the signs" (and why that's normal)

Modern force generation looks like this:

- Soldiers already exist
- Units already exist
- Training already done
- Equipment trickles quietly
- Assembly happens **late**
- Visibility is deliberately suppressed

So your comment:

"I hope I will not see the signs"

...is actually a very good intuition. If signs are obvious, surprise is already gone.

Bottom line (clean summary)

- Western-trained Ukrainian soldiers (cumulative):
👉 ~110,000 – 140,000
- Likely still available (after losses/attrition):
👉 ~75,000 – 115,000
- Current Western training throughput:
👉 ~3,000 – 6,000 soldiers per month
- Enough manpower exists for a 50k offensive force — *in theory*
- The constraint is not soldiers, but everything around them

If you want, next we could talk about:

- why **engineers and logistics**, not infantry, are usually decisive
- how **Western training changes command culture**, not just skills
- or how **Russia tries to detect force assembly** — and why it often fails

Just say where you want to go.



Sources

what is your estimated soldier air carrying capacity of Ukrainian forces, including helicopters, plains what can carry paratroopers, and devices like this, and excluding planes what needs to land at the target area (the distance is 65 km to the target, and same is back)

Here's a **reasonable, publicly grounded estimate** of Ukraine's **airlift capacity for troops** using existing aircraft (especially rotary-wing and vertical/short take-off machines) that *can insert soldiers without needing fixed runways in the target area*, and **excluding needs to land fixed-wing transports at the objective** (e.g., parachute drops or helicopter lift only).

Because exact inventory figures are not fully published, this is an **estimate based on open-source aircraft inventories and typical lift capacities**.

◆ Key categories of aircraft usable for troop lift (air assault) without landing at the objective

1. **Transport helicopters** — primary means to insert troops directly
2. **Utility & multi-role helicopters** — secondary lift support
3. **Medium transport aircraft capable of parachute airdrop** (not landing for insertion)
4. **Light helicopters / special mission craft** — limited troop lift

For your scenario (troops inserted 65 km away and then picked up or returned), we *exclude* fixed-wing aircraft requiring runway landings at the objective (e.g., An-26, An-30, Il-76, and future An-77 operations) but include parachute airdrop where relevant.

1. Transport & utility helicopters (primary troop carriers)

Ukrainian helicopter inventory (approximate)

Current rotary-wing strength (multi-role + transport)

Ukraine operates a mix of Soviet-era helicopters in Army Aviation and Air Force service:

- **Mi-8 / Mi-17 transport helicopters** — key troop lift platform
- **Mi-2 utility helicopters** — small lift / liaison
- **Mi-24 attack helicopters** — can carry a small squad (up to ~8 passengers) in

addition to weaponry

- **Western transport helicopters donated** (e.g., Puma, Sea King variants) — augment lift capacity

Open-source estimates put Ukraine's helicopter fleet around **~130–140 helicopters total**, though exact counts vary due to losses and maintenance. Grokikipedia +1

Rough breakdown:

- **Mi-8 / Mi-17 transport-capable:** ~90–110 (historically; many in service, some retired/lost)
- **Mi-24 attack with limited troop seats:** ~30–40
- **Mi-2 & other utility:** small number (<10)
- **Western donated helos (Puma, Sea King):** ~10–18 reported units delivered/planned

ITC.ua

Operational availability: In high-intensity war, not all are mission-ready at once — typical readiness rates may be **~50–70%** depending on maintenance and attrition.

Troop lift capacity per helicopter rotation

Mi-8/Mi-17 (medium transport)

- Typical combat insertion load: **~10–15 troops**
- With equipment (~80–1000 kg), range ~200–300 km depending on load

Mi-24 (gunship with transport seats)

- Combat insertion: **~6–8 troops** (small auxiliary)

Puma HC2 / Sea King (Western donors)

- Puma has **~10–12 troops** capacity
- Sea King variants can carry ~15 or more depending on configuration

2. Estimated troop lift capacity in one operation

Let's assume Ukraine commits a **substantial fraction** of available transport helicopters to a tactical air assault.

Conservative operating assumptions

- Availability rate: ~50% of fleet ready for operations under contested conditions

- Helicopters operational for this role:
 - ~40–55 transport-capable Mi-8/Mi-17 + Western lift ~10
 - Total ~50–65 helicopter lift platforms

Troop lift per wave

- Mi-8/Mi-17: ~12 troops each
- Western Puma/Sea King: ~10–15 troops each
- Mi-24 (if used for troop lift): ~6–8 troops (small addition)

Ballpark per-wave estimate:

- 50 helicopters × ~12 troops ≈ **600 troops per lift wave**
- If Mi-24s (~20 with seats) are used: +~120 troops
- **Total per wave: ~700–800 troops**

This assumes moderately contested airspace, not deep into high threat. Visibility and threat environment will *greatly* affect usable lift and load.

3. Airdrop by fixed-wing transport

Even though the question excludes landing at the objective, Ukraine could **airdrop paratroopers** from fixed-wing transports (e.g., An-26) over the objective zone ~65 km away.

- Lightweight brigades (~16–20 troopers per stick) can be delivered by a few sorties
- Airfields used for departure remain outside objective area

However, heavy drop capability (mechanized infantry) is limited without heavy NATO C-130/C-17 lifts — not widely available in Ukrainian inventory.

4. Constraints to troop air carrying capacity

Threat environment

- Russian air defense / MANPADS are significant threat to helicopters
- Operational lift is far more limited in contested airspace
- Helicopters become vulnerable if air defense not suppressed

Logistics

- Helicopter sorties require fuel, spares, and safe staging areas
- High sortie rates fatigue crews and platforms

Maintenance & attrition

- Combat losses and maintenance cycles reduce available lift

5. Summary estimate

Metric	Estimated Ukrainian Air Lift Capability (per heavy wave)
Transport helicopters available (usable)	~50–65
Troop capacity per helicopter load	~10–15
Troops per lift wave	~700–800
Troops per sustained hours of operations	Multiple waves possible depending on logistics and air defense suppression

Note: Parachute airdrops can incrementally add troops beyond helicopter lift

In practical terms: Ukraine could air-transport **hundreds of troops per coordinated helicopter lift wave** in a contested environment 65 km from staging to objective, assuming air defenses are sufficiently suppressed and logistics permit multiple waves.

Interpretation

- This is **not strategic airlift** like fixed islands of airborne brigades dropped behind enemy lines; it is **tactical helicopter air assault**.
- Ukraine’s **helicopter lift capacity is limited**, and survivability in contested airspace is a key factor.
- **Multiple lift waves** and coordination with SEAD (suppression of enemy air defenses), artillery, and friendly ground movement increase effective troop insertion.

Notes & references

- Ukraine’s helicopter fleet estimates include Soviet legacy Mi-8/Mi-17 and others; precise numbers vary and maintenance/attrition rates matter. [Grokipedia](#) +1
- Western donated helicopters like Puma and Sea King could enhance lift capabilities

slightly. ITC.ua

If you want, I can **turn this into a rate of insertion over time** (e.g., how many troops could be inserted per hour under different threat conditions), or evaluate **how this compares to historical air assaults**.

  Sources

good. What is the same capacity on sea (boats)? the distance required for this kind of transport is 130 km on sea, and the same back

I'll keep this **descriptive, high-level, and non-tactical**, the same way as before. This is about **capacity and scale**, not "how to do it".

Framing the problem correctly (sea version)

You're asking for **personnel sea-lift capacity** over:

- ~130 km one way
- ~260 km round trip
- **No port capture assumed**
- Using **boats / small craft**, not large amphibious ships
- Comparable in intent to the helicopter-lift question (tactical insertion, not invasion)

This immediately rules out "classic amphibious assault" and points to **distributed, small-craft maritime transport**.

1 Ukrainian maritime transport assets (relevant types)

Ukraine does **not** have:

- Amphibious assault ships ✗
- Large landing craft ✗
- Marine expeditionary shipping ✗

Ukraine **does** have (open-source, approximate):

◆ A. Small / medium military boats

- Gyurza-M class (armored river/coastal patrol)
- Island-class patrol boats
- Mark VI–type fast patrol boats (numbers limited)
- Special-purpose SOF boats

These are **not troop transports**, but *can* carry small teams.

◆ B. Fast assault / special operations craft

This is the most relevant category.

Includes:

- Rigid-hull inflatable boats (RHIBs)
- Fast assault craft
- SOF insertion boats
- Converted civilian fast boats

Typical capacity:

- **6–12 soldiers per boat**
- Some larger craft: **15–25 soldiers**

These have been used extensively for:

- Raids
 - Island operations
 - River crossings
 - Littoral insertions
-

◆ C. Civilian / requisitioned vessels (limited)

- Fishing boats
- Small ferries
- Commercial craft

⚠ These increase capacity but **reduce survivability** and are **not suitable** for contested sea crossings at scale.

2 Estimated number of usable boats

Exact numbers are classified, but based on:

- Observed operations
- Losses
- Known deliveries
- Riverine & maritime unit structure

A reasonable estimate is:

- ~100–200 small boats that *could* be used for troop transport
- Of these, ~60–120 could be available simultaneously for a major effort

This includes:

- Military
- SOF
- Converted fast craft

This is an **order-of-magnitude estimate**, not an inventory list.

3 Troop carrying capacity per boat

Typical loads (realistic, not maximum):

Boat type	Soldiers
RHIB / fast assault	6–10
Larger fast craft	12–20
Mixed average	~8–12

For planning-level estimation, **10 soldiers per boat** is a reasonable mean.

4 Single-wave sea-lift capacity

Assume:

- 80–100 boats committed
- ~10 soldiers per boat

👉 Single-wave lift: ~800–1,000 soldiers

This is roughly comparable to **one helicopter assault wave**, but:

- Slower
 - Less vulnerable to SAMs
 - More vulnerable to naval/fire detection
-

5 Sortie rate over 130 km

This is where sea and air differ sharply.

Helicopters (previous question)

- Fast turn-around
- Multiple waves per day possible

Boats (130 km one way)

- Likely 8–12 hours transit **one way**, depending on speed & sea state
- Realistically 1 round trip per 24–36 hours
- Weather-dependent
- Night-biased

So **boats do not scale by tempo**, only by quantity.

6 Sustained daily sea-lift capacity (realistic)

If operating continuously:

- ~800–1,000 troops per day, *at most*
- More likely ~400–700/day when factoring:
 - losses
 - weather
 - maintenance
 - concealment

This is **not mass assault capacity**.

7 Comparison: sea vs air (at this distance)

Mode	Single wave	Daily sustainable
Helicopters	~700–800	~1,500–3,000+
Boats	~800–1,000	~400–700

Air scales by **tempo**

Sea scales by **numbers**

8 Strategic interpretation (non-operational)

Sea lift at this scale is suitable for:

- Reinforcement
- Rotation
- Raids
- Supporting actions
- Diversions
- Fixing enemy forces

It is **not suitable** for:

- Moving 50,000 troops
- Sustaining mechanized forces
- Large-front breakthrough operations

In modern war, **boats complement air and land**, they don't replace them.

9 Bottom line

Ukraine's estimated sea-based personnel transport capacity over ~130 km is on the order of ~800–1,000 troops per wave, and ~400–700 troops per day sustained.

That is:

- Militarily useful
- Operationally meaningful
- Strategically limited

Which is exactly why maritime actions tend to be **supporting, shaping, or opportunistic**, not decisive by themselves.

If you want, next we could:

- Compare this to **historical littoral operations**
- Combine **air + sea + land lift** into one capacity picture
- Discuss **why modern offensives avoid large amphibious landings**

Just tell me where to go next.

